

Effect of two types of soy milk and dairy milk on plasma lipids in hypercholesterolemic adults: a randomized trial.



OBJECTIVE: To compare the effects of two commercially available soy milks (one made using whole soy beans, the other using soy protein isolate) with low-fat dairy milk on plasma lipid, insulin, and glucose responses. DESIGN: Randomized clinical trial, cross-over design. SUBJECTS: Participants were 30-65 years of age, $n = 28$, with pre-study LDL-cholesterol (LDL-C) concentrations of 160-220 mg/dL, not on lipid lowering medications, and with an overall Framingham risk score of $\leq 10\%$. INTERVENTION: Participants were required to consume sufficient milk to provide 25 g protein/d from each source. The protocol included three 4-week treatment phases, each separated from the next by a wash-out period of ≥ 4 weeks. RESULTS: Mean LDL-C concentration at the end of each phase ($\pm$ SD) was 161 $\pm$ 20, 161 $\pm$ 26 and 170 $\pm$ 24 mg/dL for the whole bean soy milk, the soy protein isolate milk, and the dairy milk, respectively ($p = 0.9$ between soy milks, $p = 0.02$ for each soy milk vs. dairy milk). No significant differences by type of milk were observed for HDL-cholesterol, triacylglycerols, insulin, or glucose. CONCLUSION: A 25 g dose of daily soy protein from soy milk led to a modest 5% lowering of LDL-C relative to dairy milk among adults with elevated LDL-C. The effect did not differ by type of soy milk and neither soy milk significantly affected other lipid variables, insulin or glucose.